Huawei Access Node BAS-ACC-37 Standard Operating Procedure (SOP) for QoS / Congestion Event Failure

Node: BAS-ACC-37 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_6853

Description

The QoS / Congestion Event failure on Huawei Access Node BAS-ACC-37 at Basavanagudi POP is attributed to the occurrence of microbursts, interface tail drops, and QoS queue starvation. This phenomenon is rooted in the complex interplay between the node's ingress and egress queues, the scheduling algorithms employed by the device, and the configuration of Quality of Service (QoS) policies. Specifically, the failure is linked to the vendor's proprietary hardware architecture, where the Broadband Universal Service Adapter (BBSU) and the Digital Signal Processor (DSP) collaborate to manage packet forwarding and queuing. When the node's processing capabilities are overwhelmed by a sudden surge in traffic, the BBSU and DSP may become desynchronized, resulting in the accumulation of packets in the ingress queue and subsequent tail drops. Furthermore, the QoS policies configured on the node may not be optimized for the specific traffic patterns, exacerbating the congestion event. Therefore, a comprehensive understanding of the Huawei Access Node's architecture and the intricacies of QoS policy configuration is essential to resolving this failure domain.

Pre-requisite Checks

- Verify that the Basavanagudi POP is accessible and that remote hands are dispatched to the site.

- Confirm that the redundant link status is operational and that there are no physical issues with the node's connections.

- Ensure that the Huawei Access Node BAS-ACC-37 is powered on and that the device is responding to ping and other basic tests.

Potential Root Causes

Cause	Probability
Microbursts causing excessive packet accumulation in the ingress queue	High
Inadequate QoS policy configuration leading to queue starvation	Medium
Hardware desynchronization between BBSU and DSP	Low

Diagnostic Commands & Expected Outputs

```
Command: display queue statistics, Expected Output: Queue statistics for interface GigabitEthernet0/0/0, Description: This command will provide insight into the current queue occupancy, packet drop rates, and scheduling algorithm utilization, helping to identify potential congestion points.
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```
Command: display qos policy all, Expected Output: QoS policy configuration for all interfaces, Description: This command will display the QoS policies configured on the node, enabling the analysis of policy settings and their potential impact on traffic forwarding.

Command: display bbsu dsp status, Expected Output: BBSU and DSP status for node BAS-ACC-37, Description: This command will provide information on the current status of the BBSU and DSP, including any errors or alarms, which may indicate hardware desynchronization.
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Resolution Steps

Step 1: Step: Optimize QoS policy configuration, Command: config qos policy priority , Description: Adjust the QoS policy configuration to ensure that the node's queues are properly provisioned and that traffic is forwarded according to the defined policies.

Step 2: Step: Adjust scheduling algorithm parameters, Command: config queue scheduling algorithm , Description: Tune the scheduling algorithm parameters to optimize queue utilization and reduce congestion.

Step 3: Step: Verify BBSU and DSP synchronization, Command: display bbsu dsp status, Description: Verify that the BBSU and DSP are synchronized and functioning correctly, and apply any necessary corrections to resolve hardware desynchronization.

Post-Check Verification

```
Command: display queue statistics, Expected Output: Queue statistics indicating reduced congestion and improved packet forwarding, Description: Verify that the queue occupancy and packet drop rates have improved after implementing the resolution steps.

Command: display qos policy all, Expected Output: QoS policy configuration indicating optimal settings for traffic forwarding, Description: Verify that the QoS policy configuration is optimal for traffic forwarding and that the node is operating within the defined policies.

Command: display bbsu dsp status, Expected Output: BBSU and DSP status indicating synchronization and correct operation, Description: Verify that the BBSU and DSP are synchronized and functioning correctly, ensuring that the node is operating within normal parameters.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: undo config qos policy , Description: Restore the original QoS policy configuration to ensure that the node operates within the defined policies.

- Step: Revert scheduling algorithm parameters, Command: undo config queue scheduling algorithm , Description: Revert the scheduling algorithm parameters to their original settings to ensure that the node operates within the expected parameters.

- Step: Verify BBSU and DSP synchronization, Command: display bbsu dsp status, Description: Verify that the BBSU and DSP are synchronized and functioning correctly after the rollback procedure.